

Python classification

K Arnold

Example Dataset: Titanic Passengers

- <https://www.openml.org/d/40945>
- <http://biostat.mc.vanderbilt.edu/wiki/pub/Main/DataSets/titanic.html>
- <http://biostat.mc.vanderbilt.edu/wiki/pub/Main/DataSets/titanic3info.txt>
- <https://www.encyclopedia-titanica.org/>

Download the data, if we don't have it already:

```
data_filename <- "data/titanic.csv"
if (!file.exists(data_filename)) {
  dir.create("data")
  download.file("https://www.openml.org/data/get_csv/16826755/phpMYEkMl", data_filename)
}
```

(r)

Python Setup

```
library(reticulate)                                     (r)  
py_config()
```

```
## python:          /Users/ka37/Library/r-miniconda/envs/r-reticulate/bin/python  
## libpython:       /Users/ka37/Library/r-miniconda/envs/r-reticulate/lib/libpython3.6m.dylib  
## pythonhome:      /Users/ka37/Library/r-miniconda/envs/r-reticulate:/Users/ka37/Library/r-miniconda/envs/  
## version:         3.6.11 | packaged by conda-forge | (default, Aug  5 2020, 20:19:23) [GCC Clang 10.0.1  
## numpy:           /Users/ka37/Library/r-miniconda/envs/r-reticulate/lib/python3.6/site-packages/numpy  
## numpy_version:   1.19.1
```

The Python Data Science Toolbox

- **Pandas** (**pd**): the main library for wrangling tabular data in Python. (analogous to *tidyverse*)
- **NumPy** (**np**): the underlying math library. Gives us **arrays** of numbers. Conventionally imported as **np**.
- **scikit-learn**: the main library for machine learning in Python.

```
import pandas as pd
import numpy as np
```

(py)

Pandas

Loading data

Data frames in R are automatically converted into Pandas **DataFrames**:

```
titanic <- read_csv("data/titanic.csv", na = "?") (r)
```

```
r.titanic.__class__ (py)
```

```
## <class 'pandas.core.frame.DataFrame'>
```

Pandas can read CSV files itself. (CSV is such a quirky data format, so read the docs for all the parameters you can set.)

```
titanic = pd.read_csv("data/titanic.csv", na_values="?") (py)
```

Exploring data structure

```
titanic.shape
```

```
(py)
```

```
## (1309, 14)
```

```
num_people, num_variables = titanic.shape  
print(f"{num_people} people, {num_variables} variables about each")
```

```
(py)
```

```
## 1309 people, 14 variables about each
```

```
titanic.head()
```

```
(py)
```

```
##      pclass  survived  ...      body      home.dest  
## 0         1         1  ...      NaN      St Louis, MO  
## 1         1         1  ...      NaN  Montreal, PQ / Chesterville, ON  
## 2         1         0  ...      NaN  Montreal, PQ / Chesterville, ON  
## 3         1         0  ...    135.0  Montreal, PQ / Chesterville, ON  
## 4         1         0  ...      NaN  Montreal, PQ / Chesterville, ON  
##  
## [5 rows x 14 columns]
```



```
titanic.info()
```

(py)

```
## <class 'pandas.core.frame.DataFrame'>
## RangeIndex: 1309 entries, 0 to 1308
## Data columns (total 14 columns):
##  #   Column      Non-Null Count  Dtype
## ---  -
##  0   pclass      1309 non-null   int64
##  1   survived    1309 non-null   int64
##  2   name        1309 non-null   object
##  3   sex         1309 non-null   object
##  4   age         1046 non-null   float64
##  5   sibsp       1309 non-null   int64
##  6   parch       1309 non-null   int64
##  7   ticket      1309 non-null   object
##  8   fare        1308 non-null   float64
##  9   cabin       295 non-null    object
## 10   embarked    1307 non-null   object
## 11   boat        486 non-null    object
## 12   body        121 non-null    float64
## 13   home.dest    745 non-null    object
## dtypes: float64(3), int64(4), object(7)
## memory usage: 143.3+ KB
```

```
titanic.describe()
```

```
(py)
```

```
##          pclass    survived    ...          fare          body
## count  1309.000000  1309.000000  ...  1308.000000  121.000000
## mean    2.294882    0.381971  ...   33.295479  160.809917
## std     0.837836    0.486055  ...   51.758668   97.696922
## min     1.000000    0.000000  ...    0.000000    1.000000
## 25%     2.000000    0.000000  ...    7.895800   72.000000
## 50%     3.000000    0.000000  ...   14.454200  155.000000
## 75%     3.000000    1.000000  ...   31.275000  256.000000
## max     3.000000    1.000000  ...  512.329200  328.000000
##
## [8 rows x 7 columns]
```

Tidying data

Drop unneeded columns

```
titanic2 = titanic.drop(['ticket', 'body'], axis = 1) (py)
```

Rename columns

```
titanic3 = titanic2.rename(columns={  
    "pclass": "passenger_class",  
    "survival": "survived",  
    "sibsp": "num_siblings_or_spouses_aboard",  
    "parch": "num_parents_or_children_aboard",  
    "ticket": "ticket_num",  
    "embarked": "embarked_from_port",  
    "boat": "lifeboat",  
}) (py)
```

Note that most (but not all!) Pandas methods make a *new DataFrame* (they don't modify the existing one).

Dropping missing data

Be careful about dropping missing data if you don't know why it's missing!

```
titanic4 = titanic3.dropna(subset = ['age', 'fare', 'embarked_from_port']) (py)
```

Querying data

Each column of a `pd.DataFrame` is a `pd.Series`, which is a NumPy array with (optional) labels.

```
titanic4['passenger_class'] (py)
```

```
## 0      1
## 1      1
## 2      1
## 3      1
## 4      1
##      ..
## 1301    3
## 1304    3
## 1306    3
## 1307    3
## 1308    3
## Name: passenger_class, Length: 1043, dtype: int64
```

You can use Boolean operations on a `Series` to get another `Series`:

```
titanic4['passenger_class'] == 1 (py)
```

```
## 0      True
## 1      True
## 2      True
## 3      True
## 4      True
##      ...
## 1301   False
## 1304   False
## 1306   False
## 1307   False
## 1308   False
## Name: passenger_class, Length: 1043, dtype: bool
```

Filtering data

You can use a Boolean series to query data:

```
is_first_class = titanic3['passenger_class'] == 1
titanic4[is_first_class]
```

```
##      passenger_class  survived  ...  lifeboat
## 0                1          1  ...          2
## 1                1          1  ...         11
## 2                1          0  ...         NaN
## 3                1          0  ...         NaN
## 4                1          0  ...         NaN
## ..              ...          ...  ...         ...
## 316              1          0  ...         NaN
## 317              1          1  ...          A
## 319              1          1  ...          3
## 321              1          0  ...         NaN
## 322              1          1  ...          8
##
## [282 rows x 12 columns]
##
## /Users/ka37/Library/r-miniconda/envs/r-reticulate/bin/python:1: UserWarning: Boolean Series key will be
```

You can combine queries too:

```
had_companions = titanic3['num_siblings_or_sons'] > 0
titanic4[is_first_class & had_companions]
```

```
##      passenger_class  survived  ...  lifeboat
## 1                1          1  ...         11
## 2                home.dest    0  ...         NaN
## 3                St Louis, MO  0  ...         NaN
## 4                Montreal, PQ / Chesterville, ON  0  ...         NaN
## 5                Montreal, PQ / Chesterville, ON  1  ...         10
## 6                Montreal, PQ / Chesterville, ON  ...  ...         ...
## 7                Montreal, PQ / Chesterville, ON  1  ...          5
## 8                Montreal, PQ / Chesterville, ON  1  ...         NaN
## 9                Switzerland / Radnor, PA        1  ...          8
## 10               Switzerland / Radnor, PA        0  ...         NaN
## 11               New York, NY / Washingtons]DC    1  ...          4
## 12               Halifax, NS
## 13               New York, NY / Washingtons]DC
##
## /Users/ka37/Library/r-miniconda/envs/r-reticulate/
```

Counting

You can get the counts of how many times each item occurs in a **Series**:

```
titanic4['passenger_class'].value_counts()
```

(py)

```
## 3      500  
## 1      282  
## 2      261  
## Name: passenger_class, dtype: int64
```


Separating data into features and outcomes

sklearn needs the features to be in a separate data frame from the outcomes, so we need to split them apart ourselves. If we want to predict survival, we can create **y** as:

```
y = titanic4['survived'] == 1 (py)
```

To create **X**, we can either drop the columns we don't want (see "tidying data" above) or directly ask for a list of columns we do want:

```
#titanic3.columns (py)
numeric_features = [
    'age', 'num_siblings_or_spouses_aborad', 'num_parents_or_children_aborad',
]
categorical_features = ['passenger_class', 'sex', 'embarked_from_port']
X = titanic4[numeric_features]
```

X.info()

(py)

```
## <class 'pandas.core.frame.DataFrame'>
## Int64Index: 1043 entries, 0 to 1308
## Data columns (total 3 columns):
##  #   Column                                Non-Null Count  Dtype
## ---  -
##  0   age                                1043 non-null   float64
##  1   num_siblings_or_spouses_aborboard  1043 non-null   int64
##  2   num_parents_or_children_aborboard  1043 non-null   int64
## dtypes: float64(1), int64(2)
## memory usage: 32.6 KB
```

Scikit-Learn (sklearn)

The documentation is very well structured:

- the **User Guide** gives narrative documentation with background and examples (e.g., **logistic regression**)
- the **API Reference** gives the nitty-gritty details about individual classes and functions
- the **Examples** show worked examples of using most components.

It's conventional to import only what you actually need from **sklearn**:

```
from sklearn.linear_model import LogisticRegression  
from sklearn.model_selection import train_test_split, cross_validate  
from sklearn.compose import make_column_transformer  
from sklearn.pipeline import make_pipeline  
from sklearn.preprocessing import StandardScaler, OneHotEncoder  
from sklearn.metrics import accuracy_score, precision_score, recall_score
```

(py)

Classifier API

All classifiers have the same basic interface: `construct`, `fit`, and `predict`.

```
clf = LogisticRegression()  
clf.fit(X, y);  
y_pred = clf.predict(X)
```

(py)

Metrics

The `sklearn.metrics` module implements a variety of useful metrics.

```
accuracy_score(y_true = y, y_pred = y_pred) (py)
```

```
## 0.6222435282837967
```

```
precision_score(y_true = y, y_pred = y_pred) (py)
```

```
## 0.6631578947368421
```

```
recall_score(y_true = y, y_pred = y_pred) (py)
```

```
## 0.14823529411764705
```

(**Remember:** "recall" = true positive rate = sensitivity; "precision" = positive predictive value)

Train-Test Split

```
np.random.seed(0)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.1)
X_train.shape, y_train.shape
```

(py)

```
## ((938, 3), (938,))
```

```
X_test.shape, y_test.shape
```

(py)

```
## ((105, 3), (105,))
```

Cross Validation

```
cv_results = cross_validate(clf, X_train, y_train, cv=5,
                             scoring=['accuracy', 'precision', 'recall'])
# Wrap the results in a DataFrame:
cv_results = pd.DataFrame(cv_results).reset_index()
```

We can now access this data in R.

```
py$cv_results
```

##	index	fit_time	score_time	test_accuracy	test_precision	test_recall
## 1	0	0.01731420	0.005382776	0.6010638	0.5625000	0.1168831
## 2	1	0.03605604	0.006525040	0.6382979	0.7647059	0.1688312
## 3	2	0.02280593	0.004880190	0.6223404	0.6500000	0.1688312
## 4	3	0.01609087	0.004525185	0.5935829	0.5000000	0.1578947
## 5	4	0.01153493	0.003874063	0.6256684	0.6875000	0.1447368

Column Transformers

Column transformers let us apply preprocessing steps to subsets of columns. For example, we'll scale the numeric features:

```
numeric_feature_proc = StandardScaler() (py)
```

and one-hot-encode the categorical features:

```
categorical_feature_proc = OneHotEncoder() (py)
```

And we'll apply each pre-processor to its corresponding columns:

```
preprocessor = make_column_transformer( (py)  
    (numeric_feature_proc, numeric_features),  
    (categorical_feature_proc, categorical_features),  
    remainder = 'drop')
```

Pipelines

Pipelines put several steps in sequence. Like *workflows* in `tidymodels`, we can use pipelines to say that the data should be preprocessed before running the model:

```
clf = make_pipeline(preprocessor, LogisticRegression())
```

(py)

Now we can use all of our features!

```
X = titanic4.drop(["survived"], axis = 1)
```

(py)

Redo the train-test split:

```
np.random.seed(0)  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.1)
```

(py)

Pipelines have the model API

```
clf.fit(X, y)
```

(py)

```
## Pipeline(steps=[('columntransformer',  
##                 ColumnTransformer(transformers=[('standardscaler',  
##                 StandardScaler(),  
##                 ['age',  
##                 'num_siblings_or_spouses_aborad',  
##                 'num_parents_or_children_aborad'])),  
##                 ('onehotencoder',  
##                 OneHotEncoder(),  
##                 ['passenger_class', 'sex',  
##                 'embarked_from_port']]))),  
##                 ('logisticregression', LogisticRegression()))]
```

CV with pipelines

A pipeline behaves exactly like a classifier (it has `fit` and `predict`), so we can use exactly the same code to validate it.

```
cv_results = cross_validate(clf, X_train, y_train, cv=5,
                             scoring=['accuracy', 'precision', 'recall'])
# Wrap the results in a DataFrame:
cv_results = pd.DataFrame(cv_results).reset_index()
```

We can now access this data in R.

```
py$cv_results
```

##	index	fit_time	score_time	test_accuracy	test_precision	test_recall
## 1	0	0.02512884	0.01159000	0.8138298	0.8088235	0.7142857
## 2	1	0.02816319	0.01322103	0.7925532	0.7567568	0.7272727
## 3	2	0.02606511	0.01480412	0.7553191	0.6781609	0.7662338
## 4	3	0.02754021	0.01128888	0.7433155	0.7121212	0.6184211
## 5	4	0.02917504	0.01227403	0.8074866	0.8030303	0.6973684